

TWIN COUNTY AP, VA, USA

WMO: 724107

Lat: 36.766N

Lon: 80.823W

Elev: 2693

StdP: 13.32

Time Zone: -5.00 (NAE)

Period: 95-19

WBAN: 53895

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.1	15.7	-3.8	4.9	18.6	0.6	6.3	20.8	29.8	27.2	26.6	24.2	12.2	300	0.428

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.3	86.1	72.0	83.8	71.2	81.7	70.4	75.7	82.3	74.2	80.3	72.8	78.4	6.2	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.3	136.9	78.6	72.4	132.8	77.6	71.7	129.6	76.8	41.4	82.2	39.7	80.3	38.5	78.5	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.4	20.5	18.0	5.0	89.3	4.8	3.3	1.5	91.7	-1.3	93.6	-4.0	95.5	-7.5	97.9
			3.4	76.3	4.9	2.2	-0.1	78.0	-3.0	79.3	-5.8	80.5	-9.4	82.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.5	35.0	37.7	44.7	54.5	62.5	69.4	72.4	71.5	65.6	55.9	45.4	38.7	
	DBStd	15.60	11.42	10.40	10.22	8.90	7.07	5.11	4.37	4.43	5.99	7.85	9.15	9.78	
	HDD50	1721	479	360	225	55	4	0	0	0	1	35	194	368	
	HDD65	4574	930	766	629	325	131	21	4	6	65	294	589	814	
	CDD50	3377	14	14	62	191	392	581	696	666	468	219	55	19	
	CDD65	753	0	0	1	11	54	151	235	207	82	12	0	0	
	CDH74	4271	0	0	3	66	285	845	1419	1180	420	52	1	0	
	CDH80	881	0	0	0	5	34	165	339	266	67	5	0	0	
Wind	WSAvg	6.8	9.1	9.0	8.9	8.3	6.3	5.2	4.5	4.1	4.6	6.4	7.6	8.0	
Precipitation	PrecAvg	45.0	3.9	2.8	4.1	3.5	4.1	4.6	4.5	4.2	3.9	3.0	3.1	3.3	
	PrecMax	60.9	8.4	5.7	8.3	6.0	8.1	9.2	7.6	7.5	13.0	9.4	6.6	7.6	
	PrecMin	35.1	1.1	1.0	0.9	0.8	1.3	0.9	1.9	1.3	0.6	0.0	0.5	1.2	
	PrecStd	6.7	1.9	1.2	1.7	1.3	1.9	2.2	1.4	1.6	2.6	2.2	1.7	1.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.9	65.8	72.8	80.7	83.5	88.0	90.3	88.7	84.5	79.3	71.7	65.8	
		MCWB	56.4	57.1	58.6	60.9	68.2	72.5	74.6	73.8	67.7	64.8	58.9	59.0	
	2%	DB	59.2	60.9	67.9	75.1	80.7	83.9	86.2	85.1	81.7	74.7	65.9	60.9	
		MCWB	54.3	52.3	55.2	59.6	66.9	70.6	73.1	72.3	68.4	63.4	56.7	55.6	
	5%	DB	55.1	56.7	63.6	72.3	77.3	81.5	83.7	82.3	79.1	71.7	63.0	56.8	
		MCWB	50.5	49.6	53.4	58.6	65.1	69.6	72.1	71.5	67.6	62.0	55.6	52.4	
	10%	DB	51.8	53.6	60.3	68.7	74.3	79.2	81.4	81.0	75.3	67.9	59.2	53.7	
		MCWB	47.6	47.2	51.4	57.1	64.3	68.4	71.4	71.0	66.8	61.1	53.1	50.0	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.5	59.8	61.9	66.1	71.9	75.7	77.4	77.9	73.5	70.2	64.0	61.7
			MCDB	62.8	63.7	69.3	71.9	78.9	83.4	84.4	83.8	79.1	74.5	65.6	63.9
2%		WB	55.9	55.4	58.8	63.4	69.4	73.4	76.0	75.5	71.8	67.6	60.3	57.1	
		MCDB	59.0	58.8	64.3	70.7	76.8	79.8	83.1	81.0	77.1	71.8	64.0	59.3	
5%		WB	52.2	51.4	55.9	61.3	67.8	71.9	74.7	73.8	70.1	65.0	57.5	53.9	
		MCDB	54.4	54.8	61.3	68.5	74.1	77.9	81.2	79.2	75.2	68.2	61.2	56.0	
10%		WB	47.6	47.6	53.5	59.1	66.1	70.3	73.0	72.4	68.4	62.6	54.4	50.0	
		MCDB	49.7	51.4	57.8	66.1	71.9	76.5	78.5	77.8	73.6	65.7	57.5	52.8	
Mean Daily Temperature Range	5% DB	MDBR	15.0	15.6	17.1	18.2	17.6	16.8	16.3	16.5	16.8	17.2	16.8	14.4	
		MCDBR	17.4	19.6	20.7	21.0	20.4	19.1	18.8	18.9	20.6	21.0	20.3	17.4	
		MCWBR	13.1	14.1	12.6	11.0	10.1	9.3	8.2	8.6	9.6	11.2	13.2	12.9	
	5% WB	MCDBR	15.3	17.3	17.2	17.9	17.2	17.1	16.7	16.5	16.6	15.9	16.4	15.1	
		MCWBR	13.4	15.4	12.1	10.8	9.5	9.3	8.6	8.4	9.0	10.4	13.3	13.4	
Clear-Sky Solar Irradiance	taub	0.288	0.297	0.321	0.359	0.396	0.425	0.456	0.461	0.365	0.332	0.299	0.285		
	taud	2.520	2.482	2.464	2.372	2.272	2.217	2.166	2.128	2.393	2.483	2.541	2.559		
	Ebn at Noon	290	300	301	292	282	272	263	259	281	283	283	285		
	Edh at Noon	25	29	33	38	43	45	47	48	35	29	24	23		
All-Sky Solar Radiation	RadAvg	740	949	1290	1625	1786	1933	1800	1661	1404	1126	847	637		
	RadStd	57	114	81	131	145	138	117	86	133	138	77	60		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (42 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+254	+155

Nomenclature: See separate page